Research protocol

WEANLESS study

Shorter weaning from invasive ventilation with levosimendan

Version 3.0, May 2025

Title: 'Shorter weaning from invasive ventilation with levosimendan: a randomized, double-blind, multicentre study in critically ill patients'

Protocol ID	WEANLESS
Short title	Shorter weaning from invasive ventilation with levosimendan
EU trial number	2024-518810-23-00
Protocol version	3.0
Protocol date	May 2025
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DOCUMENT HISTORY

Document	Date of version	Summary of Changes
Original protocol	30-1-2025	Not applicable
Version 2.0	10-4-2025	- Updated the insurance policy in section 15.4 - Rephrasing of some secondary endpoints. - Adding an extra exclusion criterion for patient on intermittent haemodialysis - Standardizing the weaning process across different study sites.
Version 3.0	9-5-2025	- Removed the deferred consent procedure - Rephrasing inclusion criteria

CONFIDENTIALITY STATEMENT

This document contains confidential information that must not be disclosed to anyone other than the sponsor, the investigative team, regulatory authorities, and members of the Research Ethics Committee.

TABLE OF CONTENTS

1. ABBREVIATIONS 8
2. SYNOPSIS 9
3. INTRODUCTION AND RATIONALE 12
3.1 Therapeutic condition and current treatment status 12
3.2 Clinical trial rationale 12
3.3 Mechanism of action, Drug class 13
3.4 Rationale for Dose Regimen/Dose Justification 13
4. STRUCTURED RISK ANALYSIS 14
4.1 Potential issues of concern 14
4.1.1 Level of knowledge about mechanism of action 14
4.1.2 Previous exposure of human beings 15
4.1.3 Induction of the mechanism in animals and/or *ex-vivo* 15
4.1.4 Selectivity of the mechanism 15
4.1.5 Analysis of potential effect 16
4.1.6 Pharmacokinetic considerations 16
4.1.7 Predictability of effect 16
4.1.8 Interaction with other products 16
4.1.9 Managing of effects 17
4.1.10 Study population 17
4.2 Overall synthesis of the direct risks for the research subjects 17
5. OBJECTIVES AND ENDPOINTS 18
6. STUDY PLAN AND DESIGN 20
6.1 Trial Design 20
6.2 Number of Patients 20
6.3 Overall study duration and follow-up 20
6.4 Patient participation 20
7. POPULATION 22
7.1 Population 22
7.2 Inclusion criteria 22
7.3 Exclusion criteria 22
7.4 Vulnerable populations and clinical trials in emergency situations 22
8. STUDY TREATMENTS 24
8.1 Investigational Medicinal Product(s) (IMP(s)) 24
8.1.1 Name and description of the IMP 24
8.1.2 Status of development of the IMP 24
8.1.3 Description and justification of dosage and route of administration 24
8.2 Comparator IMP(s) 25
8.2.1 Name and description of the comparator 25
8.3 Placebo 25
8.3.1 Name and description of the comparator 25
8.3.2 Status of development of the comparator 25
8.3.3 Description and justification justification of dosage and route of administration 25
8.4 Auxiliary Medicinal Product(s) (AxMP(s)) 25
8.4.1 Name and description of the AxMP 25
8.4.2 Statement on authorisation and justification unauthorised AxMP (if applicable) 25
8.4.3 Description and justification of dosage and route of administration 25

8.5Additional considerations for trials involving a medical device ... 25
8.6Additional considerations for trials involving an in-vitro diagnostic or companion diagnostic ... 26
8.7Preparation and labelling of the study treatment(s) ... 26
9. OTHER TREATMENTS AND RESTRICTIONS ... 27
9.1Concomitant therapy ... 27
9.1.1 Permitted medication(s) ... 27
9.1.2 Prohibited medication(s) ... 27
9.2Lifestyle restrictions ... 27
9.2.1 Contraception measures ... 27
9.2.2 Other requirements ... 27
10. TRACEABILITY, STORAGE, ACCOUNTABLILITY AND COMPLIANCE ... 28
10.1Traceability and storage of the study treatment(s)? ... 28
10.2Accountability of the study treatment(s) and compliance ... 28
11. STUDY ASSESSMENTS AND PROCEDURES ... 29
11.1Screening procedure ... 29
11.2Randomisation, blinding and treatment allocation. ... 30
11.3Study procedures and assessments ... 30
11.4Weaning process ... 31
11.4.1 SBT failure criteria ... 31
11.4.2 Extubation ... 32
11.4.3 Efficacy assessments ... 32
11.4.4 Safety assessments ... 33
12. STUDY DISCONTINUATION AND COMPLETION ... 35
12.1Definition End of Trial ... 35
12.2Criteria for temporary halt and early termination of the clinical trial ... 35
12.3Discontinuation/withdrawal of individual subjects ... 35
12.4Arrangements for subjects after their participation in the clinical trial ended. ... 36
13. SAFETY REPORTING ... 37
13.1 Definitions ... 37
13.1.1 Adverse events (AEs) ... 37
13.1.2 Serious adverse events (SAEs) ... 37
13.1.3 Suspected unexpected serious adverse reactions (SUSARs) ... 38
13.2Classification of adverse events ... 38
13.3Recording of AEs/SAEs/SUSARS ... 40
13.4Reporting of AEs and SAEs ... 40
13.4.1 Reporting of AEs by the investigator to the sponsor ... 40
13.4.2 Reporting of SAEs by the investigator to the sponsor ... 41
13.4.3 List of SAEs which do not require immediate reporting and procedure for reporting ... 41
13.5Follow-up of adverse events ... 42
13.6Reporting of SUSARs by the sponsor to EudraVigilance ... 42
13.7Annual safety report ... 42
13.8Unblinding procedures for safety reporting ... 42
13.9Temporary halt for reasons of subject safety ... 43
13.10Urgent safety measures and other relevant safety reporting ... 43
13.11Data Safety Monitoring Board (DSMB)/Data Monitoring Committee (DMC) ... 43
14. STATISTICAL ANALYSIS ... 44
14.1Description of statistical methods ... 44

14.2Analysis sets ... 44
14.3Participant demographics and other baseline characteristics ... 44
14.4Randomisation and blinding ... 44
14.5Sample size, trail power and level of significance used. ... 44
14.6Planned analysis ... 45
14.6.1 Analysis primary endpoint ... 45
14.6.2 Analysis secondary endpoint(s) ... 45
14.6.3 Analysis other study parameters/endpoints ... 45
14.7Interim analysis ... 46
14.8(Statistical) criteria for termination of the trial ... 46
14.9Procedure for accounting for missing, unused and spurious data ... 46
14.10Procedure for reporting any deviation(s) from the original statistical plan. ... 46
15. ETHICAL CONSIDERATIONS ... 47
15.1Declaration of Helsinki ... 47
15.2Recruitment and informed consent procedures ... 47
15.3Benefits and risks assessment, group relatedness ... 47
15.4Compensation for injury ... 48
15.5Compensation for subjects ... 48
15.6Compensation for investigators ... 48
15.7Other ethical considerations ... 48
16. ADMINISTRATIVE ASPECTS, MONITORING AND CONFIDENTIALITY ... 49
16.1Approval initial application and substantial modifications ... 49
16.2Monitoring ... 49
16.3Recording, handling, and storage of information ... 49
16.3.1 Handling of data and data protection ... 49
16.3.2 Source documents and case report forms (CRF) ... 50
16.3.3 Clinical trial master file and data archiving ... 50
16.3.4 Collection and storage of biological samples ... 50
16.4Audits and inspections and direct access to source data/documents ... 50
16.5Reporting of serious breaches ... 50
16.6Notification of the start and the end of the recruitment ... 50
16.7Temporary halt/(early) termination ... 51
16.7.1 Temporary halt/early termination for reasons not affecting the benefit-risk balance. 51
16.7.2 Temporary halt/early termination for reasons of subject safety ... 51
16.8Summary of the results ... 51
16.9Public disclosure and publication policy ... 51
17. REFERENCES ... 52

1. ABBREVIATIONS

AE	Adverse Event
AR	Adverse Reaction
ATMP	Advanced Therapy Medicinal Product
AxMP	Auxiliary Medicinal Product
CA	Competent Authority
CCMO	Centrale Commissie Mensgebonden Onderzoek (Central Committee on Research Involving Human Subjects)
CRA	Clinical Research Associate
CRF	Case Report Form
CRO	Contract Research Organisation
CT	Clinical Trial
CTA	Clinical Trial Authorisation
DSMB	Data Safety Monitoring Board
EC	Ethics Committee
e-CRF	Electronic Case Report Form
EU	European Union
ECG	Electrocardiogram
EMA	European Medicines Agency
EMG	Electromyogram
EMR	Electronic medical record
GCP	Good Clinical Practice
GDPR	General Data Protection Regulation
HFNC	High-flow nasal cannula
IB	Investigator’s Brochure
IC	Informed Consent
ICF	Informed Consent Form
ICH	International Conference on Harmonisation
ICU	Intensive care unit
IMP	Investigational Medicinal Product
IMPD	Investigational Medicinal Product Dossier
IP	Investigational product
MAP	Mean Arterial Pressure
MS	Member State
NIV	Non-invasive ventilation
PI	Principal Investigator
RSI	Reference Safety Information
SAE	Serious Adverse Event
SAR	Serious Adverse Reaction
SBT	Spontaneous Breathing Trial
SmPC	Summary of Product Characteristics
SOP	Standard Operating Procedure
SUSAR	Suspected Unexpected Serious Adverse Reaction
VAS	Visual analogue scale
VFD	Ventilator Free Days

2. SYNOPSIS

Title Shorter weaning from invasive ventilation with levosimendan: a randomized, double-blind, multicentre study in critically ill patients.

EU trial number 2024-518810-23-00

Rationale

Respiratory muscle weakness frequently develops in invasively ventilated patients and is strongly associated with adverse clinical outcome (1, 2). Respiratory muscle weakness in ventilated patients is partly explained by muscle atrophy (3), but also by impaired force generation at a given intracellular calcium concentration, i.e., a so-called reduced calcium sensitivity of contraction (4). Reduced calcium sensitivity reduces contractile efficiency, i.e., more energy is needed, and more CO2 is generated as compared to non-ICU patients. Levosimendan is a cardiac inotrope that improves calcium sensitivity of force generation of cardiac muscle and is approved for treatment of acute heart failure. Our group has convincingly demonstrated that respiratory muscles of invasively ventilated patients show impaired calcium sensitivity of contraction and that levosimendan improves calcium sensitivity in both experimental models and healthy subjects (5-7). To date, the effect of levosimendan on clinically relevant endpoints in ventilated critically ill patients has not been investigated.

Objective To assess the effect of levosimendan on the number of ventilator-free days up until day 28.

Main trial endpoints The number of ventilator free days from randomization up until day 28.

Secondary trial endpoints

- The number of ventilator-free days from randomization up until day 90.
- The number of days from randomization to successful weaning from invasive ventilation. Weaning success, defined as:
    - For intubated patients extubation without death or reintubation within the next 7 days, or discharge from the ICU without invasive ventilation within 7 days. Whichever comes first.
    - For tracheostomized patients, weaning success is defined as ventilation through tracheostomy without any invasive ventilation during 7 consecutive days, or discharged from the ICU with unassisted breathing, whichever came first (8).
- ICU mortality; Mortality at 28 days and 90 days post randomization
- Safety endpoints: cardiac dysrhythmia (hypotension, atrial fibrillation, ventricular arrythmias), change in dose of vasopressors after randomization, change in fluid balance after randomization.
- Dyspnea sensation. Dyspnea will be assessed daily at the end of the spontaneous breathing trial using the Visual Analogue Scale (VAS), as previously described (9). After extubation, dyspnea scores will be recorded daily in the morning until ICU discharge. Dyspnea sensation will be quantified as the median score from the first to the last measurement (9) and a difference of ≥1 point on the VAS will be considered clinically meaningful (10).
- EQ-5D-5L scores at baseline and at follow-up after 3 and 12 months.
- Reintubation (for endotracheally intubated patients) or reinstitution of invasive ventilation (for tracheostomized patients) within 7 days after liberation. An endotracheal intubation beyond 7 days is not considered a “reintubation” but will be classified as an intubation for new event within 90 days (8).
- ICU readmission within 90 days.

- The number of days with non-invasive respiratory support (high flow nasal canula or non-invasive ventilation) < 28 days after randomization
- The length of stay in the ICU.
- The length of stay in the hospital.
- Levosimendan and its metabolite pharmacokinetics, assessed for up until day 7.

Trial design

Investigator-initiated, multicentre, randomized double blinded clinical trial in ventilated adult patients admitted to the ICUs of participating hospitals. Patients will be randomized to either the levosimendan group or the control group. Each group will contain 125 subjects.

Trial population

Patients aged over 18 years, admitted to the ICU, ventilated for more than 48 hours and who have failed at least one spontaneous breathing trial (SBT) are eligible for inclusion.
Main exclusion criteria include patients with pre-existent neuromuscular disease, patients intubated primarily for neurological reasons, pregnancy, inability to obtain consent, contra-indications to levosimendan, a pre-existing treatment limitation decision, previous treatment with levosimendan within the last 30 days and current participation in another interventional trial that might interact with the study drug or the primary endpoint of this study.

Interventions

After randomization, subjects will either receive levosimendan or a placebo at a starting dose of 0.1µg/Kg bodyweight/min. In absence of rate limiting effects (heart rate >130/min, hypotension defined as MAP<65mmHg, infusion rate is increased after 4 hours to 0.2 µg/Kg/min for the remaining 20 hours. Levosimendan or placebo will be undistinguishable from each other. This dosage will be repeated weekly until successful weaning from invasive ventilator or until the primary endpoint is reached, with a maximum of four doses administered. In addition to the study medication, blood samples will be obtained at predefined timepoints, and dyspnea scores will be assessed daily during ICU admissions. Quality of life will be evaluated using a questionnaire administered at baseline and during follow-up at 3 and 12 months after initiating the study medications.

Ethical considerations relating to the clinical trial including the expected benefit to the individual subject or group of patients represented by the trial subjects as well as the nature and extent of burden and risks

This trial aims to investigate the effect of levosimendan on weaning duration from invasive ventilation in critically ill patients. These patients are critically ill, unable to breath independently, and thus rely on invasive ventilation for survival. Levosimendan has shown potential benefits in improving respiratory muscle function by enhancing calcium sensitivity, which may facilitate earlier and more successful weaning from invasive ventilation. We hypothesize that the levosimendan-induced earlier liberation from invasive ventilation will result in a shorter ICU stay, reduced complication risk and lowered mortality.
Study participants are invasively ventilated and critically ill, which makes them a vulnerable population. The main risks associated with the study are those related to the administration of levosimendan. While levosimendan has a favourable safety profile, it may potentially lead to cardiac arrhythmias or hypotension. Adverse risks will be mitigated through careful monitoring.
This trial includes a control arm. While there is no direct individual benefit from placebo administration, the inclusion of a control group – harmless to participants - is crucial for determining

the efficacy and safety of levosimendan. Care in the control group will be identical to that in the intervention group, apart from the study medication. The use of a placebo is ethically justified, as there is currently no established standard treatment for enhancing respiratory muscle function in invasively ventilated patients. All participants will continue to receive optimal ICU care.
The trial involves incapacitated patients who are unable to provide informed consent due to sedation and critical illness. We justify their inclusion because we can only answer the clinical important question in this unique population, and we cannot answer this question in non-ventilated individuals. Furthermore, the risks associated with the intervention are considered minimal.

To minimize these risks, we will closely monitor our subjects for cardiac events and hypotension and safety endpoints will be continuously reviewed by an independent Data Safety Monitoring Board (DSMB). The burden on caregivers is expected to be minimal, as patients are already admitted to the ICU and no additional visits next to standard ICU care will be required.

3. INTRODUCTION AND RATIONALE

3.1 Therapeutic condition and current treatment status

Acute respiratory failure is a life-threatening condition that may develop in the course of many diseases, including pneumonia, sepsis, trauma, chronic lung diseases or after surgery. Especially the elderly, patients with co-morbidities and immunocompromised patients are at risk to develop acute respiratory failure. In these patients, invasive ventilation is a lifesaving intervention. However, invasive ventilation is a double-edged sword. While being lifesaving, invasive ventilation is associated with serious complications which are directly related to the duration of the ventilation. Successful liberation of patients from invasive ventilation ("weaning"), represents a crucial step in the recovery process following severe lung failure. Unfortunately weaning can be a complicated process. The longer the duration of the weaning, the worsens patient outcomes, increases the risks of dying and off course increases the length of intensive care unit stay (11). Duration of invasive ventilation is among the strongest predictors of one-year functional outcome (12).
In addition, weaning duration impacts health care resource utilization. First, costs of ICU care are extremely high, in the Netherlands ± 3000 euro per day. Second, critical care resources are limited, even after the Covid pandemic. Therefore, reducing duration of ventilator weaning affects direct financial health care costs and the opportunity for other patients of consumption of finite critical care capacity. In conclusion, early ventilator weaning is extremely important for the benefit of the individual patient, but also for societal benefit.

Intensive care unit (ICU) acquired weakness of the respiratory muscles plays a significant role in difficult weaning from invasive ventilation (1). A disbalance in the load imposed on the respiratory muscles and capacity of the respiratory muscles results in ventilator weaning failure. In patients ventilated >24 hours, 63% developed respiratory muscle weakness at the time of weaning (1). No medical intervention has been shown effective in improving respiratory muscles function in ICU patients. This is surprising given that respiratory muscle weakness frequently develops in ventilated patients and is associated with adverse outcome. Therefore, interventions that improve respiratory muscle function in patients difficult to wean from invasive ventilation are much needed. The calcium sensitizer levosimendan has shown positive effects on the respiratory muscles in experimental models, healthy subjects, and ventilated ICU patients (5-7). The effects on weaning outcome are unknown.

3.2 Clinical trial rationale

Respiratory muscle weakness frequently develops in invasively ventilated patients and is strongly associated with adverse clinical outcome (1, 2). Respiratory muscle weakness in ventilated patients is partly explained by muscle atrophy (3), but also by impaired force generation at a given intracellular calcium concentration, i.e., a so-called reduced calcium sensitivity of contraction (4). Reduced calcium sensitivity reduces contractile efficiency i.e., more energy is needed, and more CO2 is generated as compared to non-ICU patients. Levosimendan is a cardiac inotrope that improves calcium sensitivity of force generation of cardiac muscle and is approved for treatment of acute heart failure. Our group has conducted a unique series of experiments demonstrating levosimendan improves calcium sensitivity of contraction in the diaphragm, the main muscle for respiration. First, levosimendan increased force generation of human diaphragm muscle fibers by ±30% (5). This was the very first study to demonstrate that pharmacological modulation of the troponin complex improves human diaphragm function. Second, in a double-blind placebo-controlled trial in healthy subjects, levosimendan improved in vivo contractile efficiency of the diaphragm by 21% (6).

Levosimendan also restored diaphragm contractility after loaded breathing (reversal of muscle fatigue). Third, in double-blind placebo-controlled study we demonstrated that levosimendan increased tidal volume, minute ventilation and decreased arterial CO2 in critically ill ventilated patients (7). No serious adverse events occurred. In conclusion, our group has convincingly demonstrated that respiratory muscles of invasively ventilated patients show impaired calcium sensitivity of contraction and that levosimendan improves calcium sensitivity. Given these encouraging results, we feel great responsibility towards patients and society to investigate the effect of levosimendan on duration of ventilator weaning.

3.3 Mechanism of action, Drug class

Levosimendan is registered as a calcium sensitizer (ATC C01CX08: other cardiac stimulants) to enhance cardiac contractility in patients with acute heart failure (13, 14). It was approved for clinic use in 2000 for the hemodynamic stabilization of patients with acutely decompensated chronic heart failure (15).
There are three well known pharmacological effects of levosimendan:

1. Increased cardiac contractility by calcium sensitization of troponin C.
2. Vasodilatation, through the opening of potassium channels on the sarcolemma of smooth muscle cells in the vasculature.
3. Cardio protection through the opening of mitochondrial potassium channels in the cardiomyocytes (16).

During the past years, research groups have described the use of levosimendan in a range of clinical conditions other than heart failure. We have performed a series of experiment to demonstrate the effect of levosimendan on respiratory muscle function on the respiratory muscles (5-7, 17).

3.4 Rationale for Dose Regimen/Dose Justification

Levosimendan is only available for intravenous treatment. Based on the safety data from three large clinical trials with levosimendan in patients with cardiogenic shock or acute heart failure (17-19) the following regimen was chosen: levosimendan is started at a rate of 0.1 µg/Kg bodyweight/min. Use of an initial bolus of levosimendan is generally not recommended, mostly to minimize the risk of vasodilation (15). In absence of rate limiting effects (heart rate >130/min, hypotension defined as MAP<65mmHg, infusion rate is increased after 4 hours to 0.2 µg/Kg/min for the remaining 20 hours. If the patient has rate-limiting side effects at the dose of 0.2 µg/Kg/min infusion rate is reduced to 0.1 µg/Kg/min or less or even discontinued (7, 20).

Because of its poor water solubility (21) and in order to maintain colour stability of the levosimendan solution, dosing will be capped at 95 kg maximum. This results in the use of 2 vials per patient, maintaining a fixed concentration (25 mg of levosimendan in 500 ml of glucose 5%: 0,05 mg/ml) and an equal yellow colour of the solution for all patients (see also: 4.1 preparation of placebo).

The study medication will be administered weekly until successful detubation or until the primary endpoint is reached. The primary endpoint is the number of ventilator-free days from randomization up until day 28, so a maximum of four doses will be administered.

4. STRUCTURED RISK ANALYSIS

4.1 Potential issues of concern

1. Risks related to levosimendan administration.

Levosimendan, the investigational medicinal product (IMP), is a well-established cardiac inotrope with a known safety profile. The main risks include cardiac dysrhythmia and hypotension as described in more detail in de SmPC of levosimendan.
See the SmPC (G2_SmPC Levosimendan).

2. Risks related to the control group.

To maintain the integrity of blinding in the study, the placebo must be indistinguishable from levosimendan. Since levosimendan has a characteristic yellow colour, a simple saline-based placebo (0,9% NaCl) is insufficient, as it would lack the necessary visual similarity.

To address this, the placebo will be prepared using a water-soluble vitamin concentrate, called Soluvit®; 10 ml diluted in 500 ml of glucose 5%. The use of Soluvit is common practice in intervention trials with levosimendan requiring a placebo, such as (22-25), LevoHeartShock (NCT04020263).

Soluvit is a water-soluble vitamin concentrate commonly used for intravenous administration in critically ill patients who require parenteral nutrition. It is considered very safe and is generally well tolerated. Side effects are rare, with nausea and vomiting occurring in only 0.1–1% of cases (26). Another risk is an imbalance in vitamins, since Soluvit contains a blend of multiple vitamins, and excessive administration could theoretically result in hypervitaminosis. However, this risk is negligible at the low dose used in this study.

For further information see the SmPC (G2_SmPC Soluvit).

3. Risks for blood sampling

Blood samples will be collected to monitor levosimendan pharmacokinetics. The number and frequency of blood draws via the existing arterial line will be minimized and aligned with standard ICU practices, posing minimal additional burden to the patient.

4.1.1 Level of knowledge about mechanism of action

Levosimendan is first described as a calcium sensitizer. That is administered to enhance cardiac contractility in patients with acute heart failure (13, 14). It was approved for clinic use in 2000 for the hemodynamic stabilization of patients with acutely decompensated chronic heart failure (15). It is approved in the Netherlands and in more than 40 countries worldwide.
There are three well known pharmacological effects of levosimendan:

- Increased cardiac contractility by calcium sensitization of troponin C.
- Vasodilatation, through the opening of potassium channels on the sarcolemma of smooth muscle cells in the vasculature.
- Cardio protection through the opening of mitochondrial potassium channels in the cardiomyocytes (16).

During the past years, research groups have described the use of levosimendan in a range of clinical situations other than heart failure. We have performed a series of experiments to demonstrate the roll of levosimendan in weaning from the invasive ventilation.

In healthy subjects and patients with chronic obstructive pulmonary disease (COPD) it improves respiratory muscle function through the improvement of calcium sensitization of the contractile proteins (5, 17). This was the very first study to demonstrate that pharmacological modulation of the troponin complex improves human diaphragm function by ±30%.
Secondly, in healthy subjects levosimendan improved in vivo contractile efficiency of the diaphragm by 21% (6). And third, in double-blind placebo-controlled study it is demonstrated that levosimendan increases tidal volume, minute ventilation and decreases arterial CO2 in critically ill ventilated patients (7). This is exactly the population we want to select for our randomized controlled trial.

4.1.2 Previous exposure of human beings

Since levosimendan is already approved in patients with acute heart failure in more than 40 countries, there are several major studies describing the risks and benefits of levosimendan in patients with heart failure or cardiogenic shock. In three major studies they included about 3500 patients and describe the risks and benefits (16):

- The safety of levosimendan in patients with left ventricular failure complicating an acute myocardial infarction was studied in RUSSLAN, a placebo-controlled double-blind, parallel-group, randomized study in 504 patients. They found that levosimendan at doses 0.1-0.2 microgram/kg/min did not induce hypotension or ischemia and reduced the risk of worsening heart failure and death in patients (27).
- The REVIVE studies (REVIVE I and REVIVE II) evaluated the efficacy of levosimendan on symptoms of heart failure by means of composite endpoint comprising patients' subjective symptom assessments and signs of worsening symptoms during the 5 days after starting a 24-h trial drug infusion with a loading dose during the first 10 minutes. They found a favorable symptomatic effect on the short-term clinical course of patients treated with levosimendan compared to placebo. They also found a higher risk of hypotension and cardiac arrhythmias during the infusion period of 24hours (18).
- The SURVIVE study was a double-blind, randomized study in 1327 patients with severe systolic heart failure comparing the effects of levosimendan with dobutamine on mortality. They showed no significant reduction in all-cause mortality despite an initial reduction in plasma B-type natriuretic peptide level in patients in the levosimendan group (19).

For further information, see the SmPC (G2_SmPC Levosimendan).

4.1.3 Induction of the mechanism in animals and/or *ex-vivo*

Levosimendan is evaluated on muscle fibres obtained from biopsies performed on patients with COPD and patients without COPD (5). This in vivo study showed that pharmacological modulation of the troponin complex improves human diaphragm muscle fibre contractility. Levosimendan improved calcium sensitivity of diaphragm fibres from patients with and without COPD. These results are very promising for using levosimendan as a treatment for diaphragm weakness.

4.1.4 Selectivity of the mechanism

Calcium sensitizers, such as levosimendan, enhance the interaction between calcium and the regulatory proteins of the contractile apparatus. This results in increased responsiveness of myofilaments to calcium, producing more force for the same amount of cytosolic calcium. Research shows that levosimendan boosts the contractility of human diaphragm muscle fibers (5). However, there is no evidence indicating that levosimendan affects the troponin complex in muscle fibers other than the diaphragm. Nonetheless, it is plausible that levosimendan could also enhance the contractility of other muscle fibers, given that the troponin complex is present in all muscle fibers,

not just in the diaphragm. It is unlikely that these effects do translate into clinically significant risks for patients since we did not observe any significant adverse outcomes related to muscle function beyond the diaphragm, and in heart failure patients who have received levosimendan, there have been no reports of muscle over contractility or dysfunction in other muscle groups, despite prolonged administration of the drug.

4.1.5 Analysis of potential effect

The dosage we will use in our study will be similar to the dosage used in patients with heart failure (no loading dose). This dosage is based on the safety data from three large clinical trials with levosimendan in patients with cardiogenic shock or acute heart failure (17-19).
The following regimen was chosen: levosimendan is started at a rate of 0.1 µg/Kg bodyweight/min. Use of an initial bolus of levosimendan is generally not recommended, mostly to minimize the risk of vasodilation (15). In absence of rate limiting effects (heart rate >130/min, hypotension defined as MAP<65mmHg, infusion rate is increased after 4 hours to 0.2 µg/Kg/min for the remaining 20 hours. If the patient has rate-limiting side effects at the dose of 0.2 µg/Kg/min infusion rate is reduced to 0.1 µg/Kg/min or less or even discontinued (7, 20).

4.1.6 Pharmacokinetic considerations

After administration intravenously, levosimendan has a short half-life of about 1 hour, which means that the steady-state concentrations of levosimendan is reached within 4-8 hours. Levosimendan is metabolized by conjugation into glycine and cysteine conjugates. ±5% is thereafter reduced to the active metabolite OR-1855 in the bowel. After absorption, OR-1855 is metabolized into active OR-1896. The Cmax of these OR-metabolites is reached 2 days after infusion is stopped. The elimination half-life of levosimendan is 1 hour. RO-1855 and OR-1896 have elimination half-lives of 75-80 hours. Clinical effects of levosimendan remain for 7 days after drug administration (28). Given the prolonged action of these metabolites, particularly on cardiovascular function, continuous monitoring will be maintained for an extended period after the infusion of levosimendan. If a patient’s condition stabilizes and they are transferred to a lower-care facility without continuous monitoring, additional precautions will be taken to ensure their safety. Nursing staff will conduct frequent checks on vital signs and protocols will be in place to promptly address any issues such as hypotension or arrhythmias. If necessary, patients can be quickly transferred back to the ICU.

4.1.7 Predictability of effect

Currently, no biomarkers have been identified or validated for predicting the effects of levosimendan. The most direct way to assess its impact is by evaluating changes in the neuro-mechanical efficiency of the diaphragm. This is typically measured using an esophageal balloon to determine transdiaphragmatic pressure, which is an invasive procedure. To avoid this, we assess the effectiveness of levosimendan through well-defined clinical endpoints, such as the "the number of ventilator free days from randomization up until day 28". This endpoint directly reflects the intended therapeutic effects.

4.1.8 Interaction with other products

No clinically relevant drug-drug interactions have been described for levosimendan (28).
In our study, all patients will be under continuous invasive hemodynamic monitoring (standard of care in ICU), and the ICU team will follow treatment protocols as necessary to maintain stable blood pressure. The use of other ICU medications, such as sedatives or inotropes, will be closely monitored to ensure that no significant drug-drug interactions occur.

4.1.9 Managing of effects

The primary anticipated side effect of levosimendan in this study population is hypotension, attributed to its vasodilatory properties. However, the patients enrolled in this trial are expected to be hemodynamically more stable compared to those with cardiogenic shock treated with levosimendan. Evidence from our pilot study demonstrated no adverse events related to levosimendan administration, reinforcing the safety of its use in this critically ill but more stable ICU cohort (7). Additionally, all patients will be managed in a high-acuity ICU setting with continuous hemodynamic monitoring, allowing for early detection and prompt intervention in the event of blood pressure changes. Standard ICU protocols include the use of invasive arterial lines or other intravascular monitoring devices, providing real-time cardiovascular data to guide clinical decision-making.

Should hypotension occur, management will adhere to established ICU protocols. In cases of persistent or severe hypotension, the levosimendan infusion rate will be reduced or discontinued, depending on the clinical context and severity of the response.

The study will also benefit from the expertise of trained ICU personnel, experienced in managing critically ill patients and skilled in recognizing and addressing adverse events promptly.

To further ensure safety, a Data Safety Monitoring Board (DSMB) will oversee the trial, conducting regular reviews of safety data. This independent oversight will facilitate the early identification and mitigation of any potential risks associated with levosimendan administration, maintaining a high standard of patient safety throughout the study.

4.1.10 Study population

Patients aged over 18 years, admitted to the ICU, ventilated for more than 48 hours and who have failed at least one spontaneous breathing trial (SBT) are eligible for inclusion. Extensive inclusion and exclusion criteria are described in section 7.2 and 7.3.

4.2 Overall synthesis of the direct risks for the research subjects

The administration of levosimendan in this study involves potential risks, particularly hypotension, due to its vasodilatory effects. However, the highly monitored ICU setting where the study takes place significantly mitigates these risks. Patients will be under continuous hemodynamic surveillance, allowing for the early detection and prompt management of complications.

We have established a management strategy for hypotension, which includes fluid resuscitation and the use of readily available vasopressors. The presence of trained ICU staff, skilled in managing critically ill patients, ensures adequate intervention in case of adverse effects. Additionally, the study will be overseen by a Data Safety Monitoring Board (DSMB), which will regularly review safety data to identify any emerging risks.

Previous studies have established levosimendan safety in acute heart failure patients and in the study described by Roesthuis et al (7) conducted in our specific ICU population showed no serious adverse events.
In summary, while there are inherent minimal risks in using levosimendan, the comprehensive monitoring, robust management strategies, and the oversight of the DSMB support the view that the remaining risks are acceptable, given the potential benefits for critically ill patients.

5. OBJECTIVES AND ENDPOINTS

Objective(s)	Endpoint(s)
Primary objective(s)	**Primary endpoint for the primary objective(s)**
To assess the effect of levosimendan on ventilator free days up until day 28	- The number of ventilator-free days (VFD) from randomization up until day 28. This is a composite endpoint combining both mortality and the duration of ventilation. • In case of death, the subject will be assigned a value of 0 VFD. • Day 0 will be defined as the day of randomization.
	Secondary endpoints
	- The number of ventilator-free days from randomization up until day 90. - The number of days from randomization to successful weaning from invasive ventilation. Weaning success, defined as: • For intubated patients extubation without death or reintubation within the next 7 days, or discharge from the ICU without invasive ventilation within 7 days. Whichever comes first. • For tracheostomized patients, weaning success is defined as ventilation through tracheostomy without any invasive ventilation during 7 consecutive days, or discharged from the ICU with unassisted breathing, whichever came first (8). - ICU mortality; Mortality at 28 days and 90 days post randomization - Safety endpoints: hypotension, cardiac dysrhythmia (atrial fibrillation, ventricular arrythmias), change in dose of vasopressors after randomization, change in fluid balance after randomization

Secondary objective(s), if applicable	**Endpoint(s) for secondary objective(s), if applicable**
To assess the effect of levosimendan on patient reported outcomes (PROM, e.g., Dyspnea sensation)	- Dyspnea sensation. Dyspnea will be assessed daily at the end of the spontaneous breathing trial using the Visual Analogue Scale (VAS), as previously described (9). After extubation, dyspnea scores will be recorded daily in the morning until ICU discharge. Dyspnea sensation will be quantified as the median score from the first to the last measurement (9) and a difference of ≥1 point on the VAS will be considered clinically meaningful (10). - EQ-5D-5L scores at baseline and at follow up after 3 and 12 months.
To assess the effect of levosimendan on health care consumption and such increases finite intensive care capacity.	- Reintubation (for endotracheally intubated patients) or reinstitution of invasive ventilation (for tracheostomized patients) within 7 days after liberation. An endotracheal intubation beyond 7 days is not considered a "reintubation" but will be classified as an intubation for new event(8) within 90 days. - ICU readmission within 90 days. - The number of days with non-invasive respiratory support (high flow nasal canula or non-invasive ventilation) < 28 days after randomization - The length of stay in the ICU. - The length of stay in the hospital.
Exploratory objective(s), if applicable	**Endpoint(s) for exploratory objective(s), if applicable**
To assess pharmacokinetics of levosimendan in critical ill patients.	- Levosimendan and its metabolite pharmacokinetics assessed for up until day 7.

6. STUDY PLAN AND DESIGN

6.1 Trial Design

WEANLESS is an investigator-initiated, multicenter, randomized double blinded clinical trial in ventilated adult patients admitted to the ICUs of participating hospitals.
The trial will employ a parallel group design, allowing for a direct comparison between the treatment group and a control group. This approach facilitates the objective measurement of the primary endpoint, ventilator free days up until day 28, while controlling for confounding variables.

In figure 1: a detailed flowchart of the study procedures is shown.

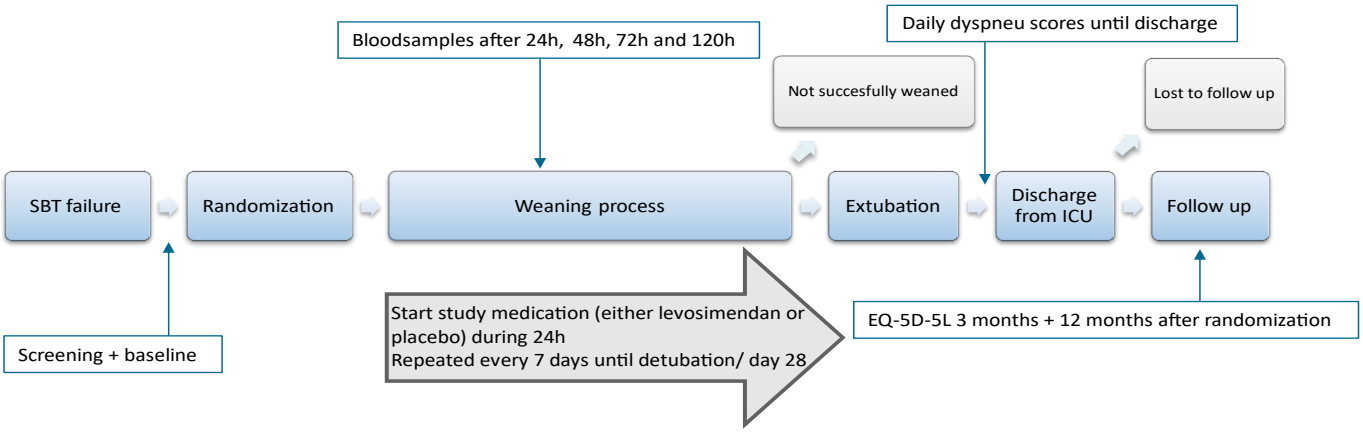

Figure 1: Overview of study procedure.

6.2 Number of Patients

We will include 250 patients in the final analysis. 125 patients in the intervention group and 125 patients in the control group. These calculations include an inflation rate of 10% to account for the possibility that ventilator free days would not be normally distributed. See section 14.5 for a detailed description.

6.3 Overall study duration and follow-up

After informed consent, the patient will be randomized and the study medication (either levosimendan or a placebo) will be administered for a 24-hour period.

The treatment can be repeated weekly until successful extubation or up to a maximum of four times. After extubation, patients will be asked to complete a brief questionnaire about their quality of life, and a follow-up questionnaire will be administered at six months.

6.4 Patient participation

We have consulted the flyer and website from “Patiëntenfederatie Nederland” and contacted mr. Jonkergauw from the Patiëntenfederatie. Given our positive experiences with patient organizations, (FCIC & European Lung Foundation (ELF)) further communication with Patiëntenfederatie is unnecessary but suggested to involve “Longfonds”. For our WEAN-SAFE project, applicants collaborated with FCIC and ELF to discuss patient relevant outcomes. Invasively ventilated patients describe dyspnea as one of their worst ICU recollections (9). Dyspnea may play a role in pathogenesis

of ICU related post-traumatic stress disorder and the impairment of quality of life (29, 30). Therefore, dyspnea is our most relevant secondary end point. A budget is allocated for involvement of patient organizations.

7. POPULATION

7.1 Population

Patients aged over 18 years, admitted to participating ICU, ventilated for more than 48 hours and who have failed at least one spontaneous breathing trial ("difficult to wean from invasive ventilation"). Participating ICUs are required to have the weaning protocol developed by the Dutch Society of Intensive Care implemented in clinical standard care. This protocol advises to perform a spontaneous breathing trial if FiO2 ≤0.50 and PEEP ≤8cmH2O, which is in line with recent literature (31-33). The spontaneous breathing trial is preferably performed without inspiratory assist and expiratory assist. Duration of spontaneous breathing trial ±60 minutes, as listed in the guideline.

7.2 Inclusion criteria

In order to be eligible to participate in this study, a subject must meet all the following criteria:

- Invasively ventilated > 48 hours.
- Failing at least one spontaneous breathing trial (SBT).
- Age above 18 years.
- Female patients of childbearing potential must have a negative pregnancy test (blood or urine) prior to participation. Post-menopausal females (> 60 years of age or no menses for 12 consecutive months without an alternative medical cause with confirmatory follicle-stimulating hormone (FSH) level of ≥ 40mIU/mL) do not require contraception during the trial.

7.3 Exclusion criteria

A potential subject who meets any of the following criteria will be excluded from participation in this study:

- Pre-existing neuromuscular disease (congenital or acquired)
- Endotracheally intubated primarily for neurological reason (e.g., traumatic brain injury, intracranial haemorrhage, epilepsy, intracranial infection), or developed severe intracranial haemorrhage/infarction during ICU stay.
- Contra-indications for levosimendan: severe renal failure (creatinine clearance <30mL/min) unless managed with appropriate continuous kidney replacement therapy (such as CRRT), severe liver failure (Child-Pugh class C), history of torsade des pointes; known significant mechanical obstructions affecting ventricular filling/ outflow or both; prolonged QTc interval (QTc > 470ms); pregnancy, breast feeding; known hypersensitivity to levosimendan.
- Treatment with intermittent haemodialysis.
- Treatment limitation decision in place: do not reintubate
- Previous treatment with levosimendan within 30 days.
- Currently in another interventional trial that might interact with study drug or primary outcome.

7.4 Vulnerable populations and clinical trials in emergency situations

The study will include invasively ventilated patients who are typically administered sedatives and analgesics, rendering them unable to provide informed consent. Excluding this significant population could lead to substantial selection bias, undermining the study's validity and generalizability. Therefore, it is essential to include these patients to accurately assess the intervention's effectiveness in a real-world clinical setting.

Most of the time a subject is incapacitated and unable to provide consent, informed consent will be sought from a legally authorized representative. This approach ensures ethical standards are upheld

while allowing the study to proceed with a population that is critical for understanding the impact of the intervention in a clinically relevant context.

8. STUDY TREATMENTS

8.1 Investigational Medicinal Product(s) (IMP(s))

8.1.1 Name and description of the IMP

Levosimendan, (-) -(R)-[[4-(1,4,5,6-tetrahydro-4-methyl-6-oxo-3-pyridazinyl) phenyl]-hydrazono] propanedinitrile, belongs to the class of calcium sensitizers. The structural formula of levosimendan is presented in Figure 1. Levosimendan is a moderately lipophilic drug with molecular weight 280.3. It is a weak acid with pKa 6.3. Solubility of levosimendan in distilled water and phosphate buffer (pH 8) is poor (0.04 mg/ml and 0.9 mg/ml, respectively). Solubility in ethanol is 7.8 mg/ml and therefore levosimendan in its pharmaceutical composition (levosimendan 2.5 mg/ ml infusion concentrate) is diluted in ethanol (34).

Levosimendan is the active pharmaceutical ingredient, and its pharmacological properties are independent of the manufacturer. In case of supply issues, switching of manufacturer will not affect the study's integrity, as efficacy and safety remain unchanged since only registered products are used.

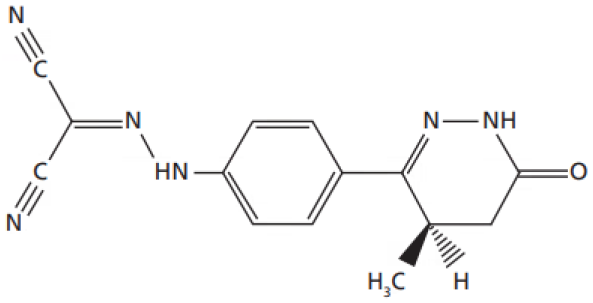

Figure 2 Structural formula of levosimendan.

8.1.2 Status of development of the IMP

Levosimendan is widely tested in humans and used in daily clinical practice for heart failure and cardiogenic shock. See the SmPC attached ('G2_SmPC Levosimendan') for further information.

8.1.3 Description and justification of dosage and route of administration

Levosimendan is currently only registered for intravenous administration. Based on safety data from three large clinical trials involving patients with cardiogenic shock or acute heart failure (17-19), we will initiate treatment at a rate of 0.1 µg/Kg bodyweight/min, avoiding a loading dose to minimize the risk of vasodilation (15). If no rate-limiting effects (heart rate >130/min or hypotension defined as MAP <65 mmHg) are observed after 4 hours, the infusion rate can be increased to 0.2 µg/Kg/min for the remaining 20 hours. Should rate-limiting side effects occur at this dose, we will reduce the infusion rate to 0.1 µg/Kg/min or discontinue it altogether (7, 20). If the patient has not been successfully weaned from invasive ventilation after 7 days, we may repeat the treatment using the same dosing schedule, up to a maximum of four cycles.

8.2 Comparator IMP(s)

8.2.1 Name and description of the comparator

Not applicable

8.3 Placebo

8.3.1 Name and description of the comparator

To maintain the integrity of blinding in the study, we will use Soluvit as a placebo since it will match the characteristic yellow colour of levosimendan. When diluted (10 mL in 500 mL glucose 5%), Soluvit® provides a yellow colour that matches the characteristic appearance of Levosimendan. This is consistent with practices in other intervention trials with levosimendan and a placebo, for example: *(22-25),* LevoHeartShock *(NCT04020263).*

Soluvit®, a sterile, water-soluble vitamin concentrate, is intended for intravenous administration. It contains a combination of essential water-soluble vitamins: thiamine (vitamin B1), riboflavin (vitamin B2), nicotinamide (vitamin B3), pyridoxine (vitamin B6), ascorbic acid (vitamin C), biotin (vitamin B7), folic acid (vitamin B9), and cyanocobalamin (vitamin B12).
The detailed composition of Soluvit® is provided in the SmPC (G2_SmPC Soluvit).

8.3.2 Status of development of the comparator

Soluvit® is a registered product widely used in clinical practice as part of parenteral nutrition to supplement essential vitamins in critically ill patients. It is well-tolerated, with an established safety profile. See the SmPC attached ('G2_SmPC Soluvit') for further information.

8.3.3 Description and justification justification of dosage and route of administration

The placebo will be prepared as a solution containing 10 mL of Soluvit® diluted in 500 ml of glucose 5% and administered intravenously. Consistent with the infusion rate of levosimendan, initial treatment will be started at a rate of 0.1 µg/Kg bodyweight/min. If no rate-limiting effects are observed after 4 hours, the infusion rate can be increased to 0.2 µg/Kg/min for the remaining 20 hours. Should rate-limiting side effects occur at this dose, we will reduce the infusion rate to 0.1 µg/Kg/min or discontinue it altogether. The route, volume and infusion rate will be matched with the administration for levosimendan maintaining the blinding in the study.

8.4 Auxiliary Medicinal Product(s) (AxMP(s))

8.4.1 Name and description of the AxMP

Not applicable.

8.4.2 Statement on authorisation and justification unauthorised AxMP (if applicable)

Not applicable.

8.4.3 Description and justification of dosage and route of administration

Not applicable.

8.5 Additional considerations for trials involving a medical device

Not applicable.

8.6 Additional considerations for trials involving an in-vitro diagnostic or companion diagnostic

Not applicable.

8.7 Preparation and labelling of the study treatment(s)

The preparation and labeling of the study treatment will strictly adhere to Good Manufacturing Practice (GMP) guidelines to ensure safety, quality, and consistency across all study sites. An additional label will be applied to the study medication, as it will be administered off-label. After labeling, the medication will be distributed to the study sites by the Radboud Trial Pharmacy.

Following randomization, qualified and unblinded nurses or pharmacy staff will prepare and label the assigned study treatment (levosimendan or placebo) in accordance with the pharmacy manual.

9. OTHER TREATMENTS AND RESTRICTIONS

9.1 Concomitant therapy

9.1.1 Permitted medication(s)

Treating physicians can prescribe any concomitant medications or treatments deemed necessary to provide adequate patient care without restrictions due to this research. Relevant medications will be recorded in case report forms, listing generic name, dose, route of administration and date of administration.

9.1.2 Prohibited medication(s)

There is no medication prohibited.

9.2 Lifestyle restrictions

9.2.1 Contraception measures

Not applicable because of ICU admission.

9.2.2 Other requirements

There are no dietary or food restrictions.

10. TRACEABILITY, STORAGE, ACCOUNTABLILITY AND COMPLIANCE

10.1 Traceability and storage of the study treatment(s)?

Levosimendan and the placebo (Soluvit®) will be ordered by the Radboud trial pharmacy. Since it will be administered off label it will receive an additional label. The charge number on the medication will be used to ensure traceability of the study medication. After labeling the medication will be distributed to the study sites and stored according to the SmPC.

Once a subject is identified for inclusion, the blinded research team will contact the designated unblinded team. For each site, it can be different who will be assigned to the unblinded team. This can be either a pharmacist at the local site or unblinded nurses (not involved in the standard ICU care). The unblinded team will obtain randomization details and prepare either levosimendan or the placebo according to the randomization, ensuring that the medication is blinded and labeled per VTGM standards to maintain blinding integrity.

Detailed documentation will be maintained at each stage of handling, from the initial storage to preparation, distribution, and final administration of the IMP. This ensures full traceability and accountability for each dose administered in compliance with regulatory standards.

10.2 Accountability of the study treatment(s) and compliance

The study medications, levosimendan and placebo, will be managed exclusively by qualified pharmacy personnel or nurses at each participating site to ensure rigorous accountability. Each site's unblinded team will also maintain a comprehensive drug accountability form, documenting the date, study identification code, charge number, and treatment allocation for each administration. Unblinded personnel will be responsible for ensuring blinding integrity, with emergency unblinding restricted to specific, safety-related situations.

Study treatments will be administered intravenously in the ICU by ICU-trained healthcare professionals, ensuring consistent compliance with the dosing regimen. This controlled administration by trained personnel guarantees subject compliance, with each dose recorded in the patient's medical record and the study-specific drug administration log, verifying adherence to the prescribed dosing schedule.

The principal investigator is responsible for maintaining an up-to-date study treatment stock sheet, including precise administration times and dates. This ensures that all study treatments are accurately tracked throughout the study, promoting consistent adherence to study protocols.

Emergency unblinding

Emergency unblinding is strictly limited to situations where knowing the patient's assigned treatment is critical for safety. In such cases, the investigator will initiate unblinding through the eCRF system (Castor), which provides secure, immediate access to the treatment assignment. This process preserves blinding integrity while allowing urgent unblinding when patient safety necessitates.

11. STUDY ASSESSMENTS AND PROCEDURES

11.1 Screening procedure

Screening and baseline

ICU patients admitted to one of the participating centres are actively screened by the treating physician to determine whether they meet the study's inclusion and exclusion criteria. Once a potential candidate is identified, informed consent will be obtained, this can happen before the patient has failed the spontaneous breathing trial (SBT). After informed consent is obtained and the patient has failed the SBT, randomization into the study will take place, followed by administration of the study medication. This step is initiated as promptly as possible to avoid unnecessary delay in starting the intervention.

Baseline assessments will be conducted after randomization but prior to administration of the study medication. All baseline procedures are detailed in Table 1.

	Pretreatment	
	Screening	**Baseline**
Inclusion/exclusion criteria	X	
Medical history	X	
Informed consent	X	
Physical examination[3]		X
Pregnancy test (urine or blood) [1]		X
Follicle stimulating hormone[2]		X
Demographics: Age, sex and ethnicity		X
APACHE II and SOFA score		X
Weight/height[3]		X
Vital signs: systolic and diastolic blood pressure, heart rate and temperature		X
12 lead ECG		X
Prior and concomitant treatment		X
Intoxications: use of alcohol, smoking status, use of drugs		X
Laboratory examinations[4]		X
Cardiac ultrasound[5]		X
Aetiology of intubation		X
Ventilation duration before study inclusion, days		X
Data from ventilator		X
EQ-5D-5L[6]		X

Table 1: Pretreatment assignments

[1] Women of childbearing potential only
[2] Only post-menopausal women
[3] If done as a standard of care and within 72 hours before IP administration, the physical examination and body weight and height do not need to be repeated at Baseline and data will be used for baseline assessment.
[4] We determine level of blood NTproBNP.
[5]If patients undergo a cardiac echocardiogram as part of standard care to evaluate potential weaning failure (35), data on ejection fraction and diastolic function will be collected. If a cardiac ultrasound is not performed as part of standard care, no additional ultrasound will be conducted for study purposes.

[6]Baseline EQ-5D-5L, is obtained either by collection the data from the MONITOR-IC database (if the patient is included in this study) or asked at the patient or their proxy to ask to fill in the questionnaire, recalling their QoL before hospital admission. We anticipate that a lot of patients are not able to fill in the questionnaire themselves, but other studies have demonstrated that proxies are able to reliably assess patients' quality of life (36-38).

11.2 Randomisation, blinding and treatment allocation.

We will use block randomization with randomly sized blocks (e.g., blocks of 6, 8 or 10 patients) to ensure balance in the number of participants allocated to each treatment group throughout the study. The randomization ratio will be 1:1 meaning participants will be equally allocated to either the intervention or control group. The randomization list will be generated using Castor, a secure, cloud-based data management system specifically designed for clinical trials, which offers robust randomization features including stratification and blocking to ensure unbiased group allocation. Castor's randomization module avoids any predictability or bias in group assignment by using validated algorithms, maintaining blinding integrity.
We apply stratified randomization to account for potential differences between the participating centers. Stratification will be based on the specific ICU site (institution) to ensure that any variability between participating centers is evenly distributed across both the intervention and control groups. This method will help mitigate confounding effects due to site-specific factors and enhance the internal validity of the study.

This study will use a double-blind approach, meaning neither the participants nor the investigators will know the treatment assignments. A matching placebo will be used to maintain blinding, ensuring that both groups receive treatments that are visually indistinguishable. Blinding will be maintained with coded study medication with the codes only being broken after data collection is complete or in the event of a medical emergency (unblinding).

11.3 Study procedures and assessments

After the patients are screened, randomized and baseline characteristics have been obtained the study medication will be administered. This will be administered within 36 hours after the failed SBT.

Treatment

<u>Study medication (levosimendan or placebo)</u>
Following randomization, patients will be assigned to either the intervention group or the control group. On the first day, they will receive the study medication, which will be administered weekly until successful detubation or until the primary endpoint is reached. The primary endpoint is the number of ventilator-free days from randomization up until day 28, with a maximum of four doses being administered.

<u>Daily dyspnea sensation score</u>
Dyspnea will be assessed daily at the end of the spontaneous breathing trial using the visual analogue scale. After detubation, dyspnea scores are assessed daily in the morning until ICU discharge.

Blood samples
Blood sampling through an existing arterial/venous cannula line to assess the pharmacokinetics from levosimendan in critical ill patients. This will be performed at 24hours, 48hours, 72hours and 120hours after completing first dose of study medication. We will also evaluate levels of NTproBNP at baseline. Blood sampling will be conducted in most patients; however, it may not be feasible in all hospitals, so not all patients will undergo these assessments.

EQ-5D-5L
The EuroQol 5-dimensions 5-level questionnaire is a standardized instrument to describe and value health, consisting out of a descriptive system and a visual analog scale (VAS). The EQ-5D-5L will be performed at baseline (describing the situation before ICU admission), after 3 months and after 12 months). If a patient is unable to complete the questionnaire, a proxy will be asked. Each questionnaire provided to a patient must contain the date of the assessment and patient number and be checked for completeness. After 3 (± 14 days) and after 12 (± 14 days) months, the questionnaire will be completed by a phone interview, or the questionnaire will be sent to the subjects by mail. Responses will be recorded in the eCRF.

Parameters assessed daily during the first 28 days or discharge from the ICU (if earlier):

- Vital signs
- When ventilated: data from ventilator. Settings, tidal volume, pressures.
- Attempt of SBT
- Use of non-invasive ventilatory support.

11.4 Weaning process

To standardize the weaning process across all study sites, physicians are required to assess daily whether patients meet the criteria for a spontaneous breathing trial (SBT). The criteria align with the weaning protocol developed by the Dutch Society of Intensive Care (39), which is supported by recent literature (31-33). We strongly encourage physicians to adhere to the criteria outlined below.

Criteria for initiating a spontaneous breathing trial (SBT):

1	Adequate oxygenation	SpO_2 > 90% met FiO_2 ≤ 50% and PEEP ≤ 8 cmH_2O
2	Adequate respiratory rate	Respiratory rate ≤ 34 /min
3	Stable cardiovascular status	Systolic blood pressure 90-160mmHg without or minimal vasopressor use, heart rate ≤ 140/min

Table 2: Criteria for SBT

When a patient meets all of the above criteria, an SBT will be performed. This trial should last at least 30 minutes and can be conducted using one of the following methods:

1. Using the T-piece
2. Pressure support mode, with PS ≤ 8 cmH2O and PEEP ≤ 7 cmH2O
3. CPAP mode, with PEEP ≤ 7 cmH2O

11.4.1 SBT failure criteria

SBT failure will be documented in the eCRF if the patient develops any of the following:

- Worsening oxygenation (SpO2 < 90% with FiO2 ≥ 50%),
- Acute distress (respiratory rate ≥ 40/min, agitation, cyanosis) or
- Hemodynamic instability (systolic blood pressure ≤ 90 or ≥ 180 mmHg, heart rate > 140/min or increase of 20%

11.4.2 Extubation

Following a successful SBT, extubation should be considered unless contraindications exist. The decision to extubate remains at the discretion of the treating intensivist. The study protocol does not mandate specific reintubation criteria but requires documentation of all reintubation events, including reasons for reintubation when applicable.

Loss to follow up

Procedures for patients who are lost to follow up or want to withdrawal from the study are described in section 12.3.

11.4.3 Efficacy assessments

Primary endpoint

1. Number of ventilator free days from randomization up until day 28.
   Timing: Ventilation status will be recorded daily from the initiation of the study medication until ICU discharge or weaning success (up until 28 days).
   Recording: Ventilator settings, extubation times, and ICU discharge dates will be logged in the electronic medical record (EMR) and study database.

Secondary endpoints

1. Number of ventilator-free days from randomization up until day 28:
   Timing: Daily, up until day 28.
   Recording: Ventilator-free days will be documented based on invasive ventilation status recorded daily in the EMR.
2. Number of ventilator-free days from endotracheal intubation up until day 28 and day 90:
   Method: Ventilator-free days will be counted from the time of intubation to either 28- or 90-days post-randomization.
   Timing: Daily, up until day 28 and day 90.
   Recording: Ventilator-free days will be logged daily in the EMR.
3. Number of days from initiation of study medication to successful weaning from invasive ventilation.
   Timing: Ventilation status will be recorded daily from the initiation of the study medication until ICU discharge or weaning success (up to 28 days).
   Recording: Ventilator settings, extubation times, and ICU discharge dates will be logged in the electronic medical record (EMR) and study database.
4. ICU mortality; mortality at 28- and 90-days post-randomization:
   Method: mortality will be tracked by reviewing patient medical records
   Timing: mortality status will be recorded at 28 days and 90 days post-randomization.
5. Dyspnea sensation:
   Method: dyspnea will be assessed using the visual analogue scale (VAS), with daily evaluations after spontaneous breathing trials (SBT) until ICU discharge.
   Timing: daily in the morning from SBT until ICU discharge.
   Recording: dyspnea scores will be entered into the study database.
6. EQ-5D-5L at baseline at during follow up.
   Method: the EQ-5D-5L questionnaire will be administered at baseline, and 3- and 12-months post-discharge.
   Timing: at baseline, and 3- and 12-months post-discharge.
   Recording: EQ-5D-5L scores will be entered into the study database.
7. Reintubation or reinstitution of invasive ventilation within 7 days after liberation:
   Method: reintubation or reinstitution of invasive ventilation will be tracked for 7 days post-liberation from invasive ventilation.

Timing: daily monitoring during the 7 days post-liberation.
Recording: instances of reintubation or reinstitution of ventilation will be recorded in the EMR.

8. ICU Readmission within 90 Days:
   Method: ICU readmission will be tracked through hospital records.
   Timing: Up to 90 days post-randomization.
   Recording: readmission dates and reasons will be recorded in the study database.
9. Non-invasive respiratory support use (HFNC/NIV):
   Method: Use of high-flow nasal cannula or non-invasive ventilation will be monitored.
   Timing: daily, up to 28 days post-randomization.
   Recording: support use will be recorded in the EMR and study database.
10. Length of ICU stay:
    Method: ICU length of stay will be determined from hospital records.
    Timing: tracked from ICU admission until discharge.
    Recording: length of stay will be recorded in days in the study database.

Exploratory endpoints

1. Levosimendan and metabolite pharmacokinetics:
   Method: blood samples will be taken to assess drug and metabolite levels.
   Timing: samples will be collected at predetermined intervals during the first 7 days.
   Recording: concentration levels will be recorded in the study database.

11.4.4 Safety assessments

1. **Adverse events**: all adverse events (AEs), serious adverse events (SAEs and suspected unexpected serious adverse reactions (SUSARs) will be documented and reported as outlined in chapter 13.

2. **Cardiac dysrhythmia**: Continuous electrocardiogram (ECG) monitoring will be used to detect dysrhythmias up to 30 hours after the initiation of study medication infusion, conform registration in the LIDO trial (17). Any dysrhythmia will be categorized by type (e.g., atrial fibrillation, ventricular arrhythmia) and analyzed in relation to timing of the study drug administration.

3. **Heart rate and blood pressure:**
   a. As part of routine ICU care, blood pressure will be monitored continuously during infusion. A deviation from baseline mean arterial pressure (MAP) by 20% or more, sustained for at least 10 minutes, will be considered significant and documented.
   b. Heart rate will also be monitored continuously, with a 20% or more change from baseline sustained for a minimum of 10 minutes deemed significant and recorded.

4. **Medication Errors and Overdose:** Medication errors and overdose must be documented on the respective eCRF page and are defined as follows:
   a. Medication error: any unintentional error in the administration of the investigational product (IP) while under the healthcare professional's control.
   b. Overdose: Administration of a quantity of the IP given per administration or cumulatively which is above the maximum dose.

   Each occurrence will be documented with details including the start and stop dates and actions taken with the IP (e.g., no change, temporary interruption, dose adjustment).
   If a medication error or overdose is accompanied by an adverse event, this AE will also be recorded on the AE page of the eCRF.

5. **Vasopressor and inotropic medication usage**: The use of vasopressors and/or inotropic drugs will be documented daily from start study medication till end of ICU stay. Generic name, start and stop date and time and the daily dose will be recorded in the eCRF.

12. STUDY DISCONTINUATION AND COMPLETION

12.1 Definition End of Trial

The end of the study is defined as the last patient's last visit.
In case the study is ended prematurely, the sponsor will notify the accredited METC and the competent authority within 15 days, including the reasons for the premature termination.

12.2 Criteria for temporary halt and early termination of the clinical trial

In accordance with section 10, subsection 4 of the WMO, the sponsor will suspend the study if there is sufficient ground that continuation of the study will jeopardize subject health or safety. The sponsor will notify the accredited medical EC without undue delay of a temporary halt including the reason for such an action. The study will be suspended pending a further positive decision by the accredited EC. The investigator will take care that all subjects are kept informed.

12.3 Discontinuation/withdrawal of individual subjects

In accordance with the Declaration of Helsinki, ICH Good Clinical Practice Guidelines and the US FDA Regulations, a patient has the right to withdraw from the study at any time for any reason without prejudice to his/her future medical care by the physician or at the institution. Shout a patient (or a patient's legally authorized representative) decide to withdraw, all efforts will be made to complete and report the observations as thoroughly as possible. The investigator can decide to withdraw a subject from the study for
a) urgent medical reasons or
b) if a protocol violation occurs and
c) if the subject is lost to follow-up.

A complete final evaluation should be made at the time of the patient's withdrawal. While an explanation for withdrawal may be documented if provided voluntarily, patients or their legal representatives are not required to disclose a reason.
If consent is not obtained or the legal representative declines participation, the patient is withdrawn, their data is discarded, and the randomization code is released back into the pool, ensuring it does not count toward the total number of randomized patients.

If a participant is lost to follow-up, every effort should be made to contact the participant's primary physician (GP) to obtain information on the participant's status. Similarly, if a patient's care is transferred to another clinician, every effort should be made so that follow-up information can be obtained.
A subject will be considered lost to follow-up if he or she is unable to be contacted by the trial site staff.
The following actions must be taken:

- Before a subject is deemed lost to follow-up, the investigator or designee will make every effort to regain contact with the subject. At least three telephone calls. These contact attempts should be documented in the subject's medical record or study file.
- Every effort should be made to contact the participant's primary physician to obtain information on the participant's status. Similarly, if a patient's care is transferred to another clinician, every effort should be made so that follow-up information be obtained.
- Should the subject continue to be unreachable, he or she will be considered to have withdrawn from the trial with a primary reason of lost to follow-up.

12.4 Arrangements for subjects after their participation in the clinical trial ended.

Not applicable.

13. SAFETY REPORTING

13.1 Definitions

13.1.1 Adverse events (AEs)

An adverse event (AE) is any untoward medical occurrence (e.g., symptom, sign, diagnosis or diagnostic test finding) in a patient enrolled into a clinical study regardless of its causal relationship to study drug. An AE can therefore be any unfavorable and unintended sign (including an abnormal laboratory finding), symptom, or disease (new or exacerbated).

Events meeting the adverse event definition:

- Any abnormal laboratory test results (haematology, clinical chemistry, or urinalysis) or other safety assessments (e.g., electrocardiogram [ECG], radiological scans, vital signs measurements), including those that worsen from baseline, considered clinically significant in the medical and scientific judgment of the Investigator (i.e., not related to progression of underlying disease).
- Exacerbation of a chronic or intermittent pre-existing condition including either an increase in frequency and/or intensity of the condition.
- New conditions detected or diagnosed after study drug administration even though it may have been present before the start of the study.
- Signs, symptoms, or the clinical sequelae of a suspected drug-drug interaction.
- Signs, symptoms, or the clinical sequelae of a suspected overdose of either study drug or a concomitant medication.

Events not meeting the adverse event definition:

- Any clinically significant abnormal laboratory findings or other abnormal safety assessments which are associated with the underlying disease, unless judged by the Investigator to be more severe than expected for the patient's condition.
- The disease/disorder being studied or expected progression, signs, or symptoms of the disease/disorder being studied, unless more severe than expected for the patient's condition.
- Medical or surgical procedure (e.g., endoscopy, appendectomy): the condition that leads to the procedure is the AE.
- Situations in which an untoward medical occurrence did not occur (social and/or convenience admission to a hospital).
- Anticipated day-to-day fluctuations of pre-existing disease(s) or condition(s) present or detected at the start of the study that do not worsen.

In the ICU setting, patients frequently experience routine fluctuations in laboratory values and symptoms as a result of their critical illness and intensive care interventions (40). These fluctuations, along with expected symptoms such as transient shifts in electrolyte levels, brief hypotensive episodes responsive to standard fluid resuscitation, or low-grade fever, are considered typical for this population and do not automatically constitute adverse events. Such occurrences will only be classified as AEs if they are judged by the Investigator to be clinically significant or outside the expected range of ICU variability. This will help distinguish clinically relevant AEs from expected ICU-related deviations, streamlining the process and focusing on significant adverse outcomes.

13.1.2 Serious adverse events (SAEs)

Serious adverse event is any untoward medical occurrence in a patient or trial subject that at any dose:

- Results in death

- Is life-threatening, the term 'life-threatening' in the definition of 'serious' refers to an event in which the patient was at risk of death at the time of the event. It does not refer to an event, which hypothetically might have caused death, if it were more severe.
- Requires inpatient hospitalization or prolongation of existing hospitalization. In general, hospitalization signifies that the patient has been detained (usually involving at least an overnight stay) at the hospital or emergency ward for observation and/or treatment that would not have been appropriate in the physician's office or outpatient setting. Complications that occur during hospitalization are AE. If a complication prolongs hospitalization or fulfils any other serious criteria, the event is serious. When in doubt as to whether "hospitalization" occurred or was necessary, the AE must be considered serious. Hospitalization for elective treatment of a pre-existing condition that did not worsen from baseline is not considered an AE.
- Results in persistent or significant disability/incapacity,
    - The term disability means a substantial disruption of a person's ability to conduct normal life functions.
    - This definition is not intended to include experiences of relatively minor medical significance such as uncomplicated headache, nausea, vomiting, diarrhea, influenza, and accidental trauma (e.g., sprained ankle) which may interfere with or prevent everyday life functions but do not constitute a substantial disruption.
- is a congenital anomaly/birth defect.

13.1.3 Suspected unexpected serious adverse reactions (SUSARs)

Unexpected adverse reactions are SUSARs if the following three conditions are met:

1. The event must be serious.
2. There must be a certain degree of probability that the event is a harmful and an undesirable reaction to the medicinal product under investigation, regardless of the administered dose.
3. The adverse reaction must be unexpected the nature and severity of the adverse reaction are not in agreement with the product information as recorded in the SmPC.
   See the SmPC (G2_SmPC Levosimendan).

13.2 Classification of adverse events

Severity

The severity of an AE refers to the extent to which an AE affects the patient's daily activities. The investigator will assess severity for each AE and SAE reported during the study and assign it to one of the following categories:

Mild	An event that is easily tolerated by the patient, causing minimal discomfort, and not interfering with everyday activities.
Moderate	An event that causes sufficient discomfort and interferes with normal everyday activities.
Severe	An event that prevents normal everyday activities. An AE that is assessed as severe must not be confused with an SAE. Severe is a category utilized for rating the intensity of an event; and both AE and SAE can be assessed as severe.

When changes in the severity of an AE occur more frequently than once a day, the maximum severity for the event must be noted for that day. Any change in severity of signs and symptoms over a number of days will be captured by recording a new AE, with the amended severity grade and the date (and time, if known) of the change. Changes in the severity of an AE must be documented to allow an assessment of the duration of the event at each level of intensity to be performed.

Causality

The Investigator is obligated to assess the relationship between study drug and each occurrence of each AE/SAE. A "reasonable possibility" of a relationship conveys that there are facts, evidence, and/or arguments to suggest a causal relationship, rather than a relationship cannot be ruled out. The investigator will use clinical judgment to determine the relationship.

- Alternative causes, such as underlying disease(s), concomitant therapy, and other risk factors, as well as the temporal relationship of the event to study drug administration will be considered and investigated.
- For each AE/SAE, the Investigator must document in the medical notes that he/she has reviewed the AE/SAE and has provided an assessment of causality.
- There may be situations in which an SAE has occurred, and the Investigator has minimal information to include in the initial report to the Sponsor or its designee. However, it is very important that the Investigator always make an assessment of causality for every event before the initial transmission of the SAE data to the Sponsor or its designee.
- The Investigator may change his/her opinion of causality in light of follow-up information and send an SAE follow-up report with the updated causality assessment.

The causal relationship between the AE and the IP and trial procedure will be assessed as follows:

Definitely related	There is a reasonable temporal relationship between the trial intervention and the AE, and no other obvious explanation for the AE exists.
Probably related	There is a reasonable temporal relationship between the trial intervention and the AE, based upon the known pharmacological action of the IP, known, or previously reported adverse reactions to the IP or class of drugs, or judgment based on the investigator's clinical experience, the association of the event with the IP seems likely. The event disappears or decreases on cessation or reduction of the dose of IP.
Possible related	There is a reasonable temporal relationship between the trial intervention and the AE, and the AE could also be explained by disease, other medications, or another obvious explanation of the AE exists.
Unrelated	The AE is independent of any trial intervention, or evidence exists that the event is definitely related to another etiology.

Outcome categories

Resolved	The patient has fully recovered from the event, or the condition has returned to the level observed at baseline.
Resolving	The patient has recovered from the event, but the condition has not returned to the level observed at baseline.
Not resolved	The event is ongoing at the time of reporting and the condition has not returned to the level observed at baseline.
Resolved with sequelae	As a result of the AE, the patient suffered persistent and significant disability/incapacity, or the condition became chronic or stabilized and no worsening is expected.
Death	The patient died due to the event. If the patient died due to other circumstances than the event, the outcome should be stated otherwise.
Unknown	If outcome is not known or not reported.

Action taken with IP

The action taken with the IP will be assigned to one of the following categories:

Dose not changed	No change in the dose of the IP.
Drug withdrawn	Discontinuation of the IP.
Unknown	The information is unknown or implausible and it cannot be supplemented or verified.
Not applicable	This category should be used in circumstances such as when the patient has died, or the treatment has been completed prior to reactions or events or the IP had not been administered.

13.3 Recording of AEs/SAEs/SUSARS

The investigator and designated, qualified personnel at each clinical site are responsible for detecting, documenting, and reporting events that meet the definition of AEs, TEAEs or SAEs regardless of their relationship to trial drug. Each AE must be assessed for seriousness, causality and severity (using severity grades defined in section 13.2).

It is recognized that the patient population in the ICU or intermediate care unit will experience a number of common aberrations in laboratory values, signs, and symptoms due to the severity of the underlying disease and the impact of standard therapies. These will not necessarily constitute an AE unless they require significant intervention, lead to discontinuation of blinded trial drug, or are considered unanticipated or more severe than expected for the patient's medical condition based on the investigator's clinical judgment. See section 13.1.

All AEs, regardless of their suspected relationship to trial drug, will be collected and assessed from the time the patient signs the ICF up to until day 28 or until ICU discharge (whichever comes first). Information to be recorded on the AE page of the eCRF includes:

- Event term
- Time of onset
- Investigator-specified assessment of severity and relationship to trial drug
- Date of resolution of the event
- Seriousness
- Any required treatment or evaluations
- Outcome

A continuous event changing in severity will be considered one event with the most severe intensity documented.

13.4 Reporting of AEs and SAEs

13.4.1 Reporting of AEs by the investigator to the sponsor

In this study, reporting of adverse events will be restricted to events that are considered to be related to study assigned administration method (possibly, probably or definitely). The rationale for this procedure is to prevent expedited reporting of events commonly occurring in this critically ill population that are unlikely to be related to the investigational product, thereby focusing on potential delayed effects associated with the study treatment. This procedure is commonly used in other trials in the ICU (41). Events collected as study outcomes will not be reported as adverse events.

Any adverse events thought to be related to study assigned administration method will be reported within 7 days of discovery. The site principal investigator will be responsible for determining the

causal relationship as either possible, probable or definitely related. Notification will be by completing an adverse event form on the web-based data management system.

13.4.2 Reporting of SAEs by the investigator to the sponsor

- The investigator must report any SAEs to the Sponsor or its designee within 24 hours after site staff first learns about the event.
- Given that critically ill patients are likely to meet the criteria for SAEs in the course of their ICU admission, only SAEs that the investigator considers possibly, probably, or definitely related to the study administration will be reported.
- When reporting an SAE the event term should ideally be the diagnosis of the event based on signs, symptoms, and/or other clinical information. Causality assessment should also be provided.
- The Sponsor or its designee will review each SAE report and the Sponsor, or its designee will evaluate the seriousness and the causal relationship of the event to study drug. In addition, the Sponsor or its designee will evaluate the expectedness according to the SmPC. Based on the Investigator and Sponsor's assessment of the event, a decision will be made concerning the need for further action. The Sponsor or its designee will be responsible for all information processing and reporting according to local legal requirements.
- All SAEs reported will be documented in a line listing to facilitate cumulative safety monitoring and periodic review for potential trends or safety signals related to the study drug.

Format of line listing:

No.	Name	Start date	End date	Severity Mild 1 Moderate 2 Severe 3	Relationship to IP Unrelated: 1 Possibly Related: 2 Probably Related: 3 Definitely Related: 4	Relationship to Study Protocol Unrelated: 1 Possibly Related: 2 Probably Related: 3 Definitely Related: 4	Actions taken None: 0 Concomitated medication: 1 Therapeutic/diagnostic procedure: 2	Action taken with IP Not Applicable: 0 Dose Not Changed: 1 Drug Interrupted: 2 Drug Withdrawn: 3 Unknown: 4	Outcome: Resolved: 1 Resolving: 2 Not Resolved / Not Resolving: 3 Resolved with Sequelae: 4 Death: 5 Unknown: 0	SEA Yes/No	Sig. and date

13.4.3 List of SAEs which do not require immediate reporting and procedure for reporting

As described in section 13.4.1, in this study, reporting of adverse events will be restricted to events that are considered to be related (possibly, probably or definitely) to the study assigned administration method. The rationale for this procedure is to prevent expedited reporting of events commonly occurring in this critically ill population.

13.5 Follow-up of adverse events

- All AEs must be reported in detail on the appropriate page of the eCRF and followed up until day 28 or ICU discharge (whichever comes first) or to satisfactory resolution, until the investigator deems the event to be chronic or not clinically significant or until the patient is considered to be stable if before day 28.
- Ongoing SAEs on day 28 will be followed to satisfactory resolution, until the investigator deems the event to be chronic or not clinically significant, until the patient is considered to be stable, or patient is lost to follow-up.
- The investigator is obligated to perform or arrange for the conduct of supplemental measurements and/or evaluations as medically indicated or as requested by the Sponsor or its designee to elucidate the nature and/or causality of the AE or SAE as fully as possible. This may include additional laboratory tests or investigations, histopathological examinations, or consultation with other health care professionals.
- New or updated information will be recorded in the originally completed eCRF.
- The investigator will submit any updated SAE data to the sponsor or its designee within 24 hours of receipt of the information.

13.6 Reporting of SUSARs by the sponsor to EudraVigilance

The sponsor will keep detailed records of all AEs which are reported to him/her by the investigator or investigators (CTR: Article 41(3)).

The sponsor will report electronically and without delay to EudraVigilance all relevant information about any SUSAR (CTR: Article 42).

The period for the reporting of SUSARs by the sponsor to EudraVigilance will take account of the seriousness of the reaction and will be as follows:

- In the case of fatal or life-threatening SUSARs, as soon as possible and in any event not later than **7 days** after the sponsor became aware of the reaction (CTR: Article 42(2(a)).
- In the case of non-fatal or non-life threatening SUSARs, not later than **15 days** after the sponsor became aware of the reaction (CTR: Article 42(2(b)).
- In the case of a SUSARs which was initially considered to be non-fatal or nonlife threatening but which turns out to be fatal or life-threatening, as soon as possible and in any event not later than **7 days** after the sponsor became aware of the reaction being fatal or life-threatening (CTR: Article 42(2(c)).

Where necessary to ensure timely reporting, the sponsor may, in accordance with section 2.4 of Annex III, submit an initial incomplete report followed up by a complete report (CTR: Article 42(2)).

13.7 Annual safety report

Regarding investigational medicinal products other than placebo, the sponsor shall submit annually through CTIS to all Member States concerned a report on the safety of each investigational medicinal product used in a clinical trial (CTR: Article 43).

13.8 Unblinding procedures for safety reporting

The investigator will only unblind the treatment allocation of a subject during a clinical trial if unblinding is relevant to the safety of the subject (CTR: Annex III 2.5(17)).

When reporting a SUSAR to the EMA, the sponsor will only unblind the treatment allocation of the affected subject to whom the SUSAR relates (CTR: Annex III 2.5(18)).

In case of unblinding, describe procedure to maintain blind for persons responsible for the ongoing conduct of the clinical trial such as the management, monitors, investigators) and those persons responsible for data analysis and interpretation of results at the conclusion of the clinical trial, such as biometrics personnel (CTR: Annex III 2.5(19)).

Unblinded information will be accessible only to persons who need to be involved in the safety reporting to the EMA, or to persons performing ongoing safety evaluations during the clinical trial (CTR: Annex III 2.5(20)).

13.9 Temporary halt for reasons of subject safety

The sponsor will suspend the study if there is sufficient ground that continuation of the study will jeopardise subject health or safety. The sponsor will submit the notification through CTIS without undue delay of a temporary halt but not later than in 15 days of the date of the temporary halt. It shall include the reasons for such action and specify follow-up measures. The study will be suspended pending a further positive decision by the concerned member state (CTR: Article 38). The investigator will take care that all subjects are kept informed.

13.10 Urgent safety measures and other relevant safety reporting

Where an unexpected event is likely to seriously affect the benefit-risk balance, the sponsor and the investigator will take appropriate urgent safety measures to protect the subjects. In addition, the sponsor will notify the Member States concerned, through CTIS, of the event and the measures taken. That notification will be made without undue delay but no later than **7 days** from the date the measures have been taken (CTR: Article 54).

13.11 Data Safety Monitoring Board (DSMB)/Data Monitoring Committee (DMC)

A Data Safety Monitoring Board (DSMB) will be established to oversee the safety of the study interventions and the overall conduct of the trial. The DSMB will consist of four independent experts who are not involved in the conduct of the trial. Their primary responsibilities will include safeguarding patient safety and monitoring the progress and integrity of the study.

In addition to evaluating the safety of participants, the DSMB will review other critical aspects of the trial, such as recruitment progress, adherence to the protocol, and any emerging issues that could impact the study's validity or feasibility. This includes assessing inclusion rates, addressing frequent protocol violations, and identifying any operational challenges that may arise during the trial. The DSMB will meet at predetermined intervals to discuss these matters and provide recommendations to ensure the study remains on track and aligned with its objectives.

The advice(s) of the DSMB will only be sent to the sponsor of the study. Should the sponsor decide not to fully implement the advice of the DSMB, the sponsor will send the advice to the reviewing METC, including a note to substantiate why (part of) the advice of the DSMB will not be followed.

Details of the DSMB’s responsibilities and procedures can be found in the DSMB charter.

14. STATISTICAL ANALYSIS

14.1 Description of statistical methods

All statistical analysis will be conducted with R (http://www.r-project.org/). A p-value of < 0.05 is considered significant. Baseline characteristics of the study population will be summarized using descriptive statics. Continuous variables will be presented as means and standard deviations.

14.2 Analysis sets

We will perform an intention to treat analysis which means participants will be analysed according to the group they were originally assigned to at randomization regardless of whether they completed the study protocol as planned.

14.3 Participant demographics and other baseline characteristics

Patients' demographics and baseline characteristics will be summarized for each group at the time of randomization. This will provide an overview of the comparability between the groups and help identify any imbalances that may impact the outcomes.

14.4 Randomisation and blinding

We will use block randomization with random block sizes to ensure balance in the number of participants allocated to each treatment group throughout the study. The randomization ratio will be 1:1 meaning participants will be equally allocated to either the intervention or control group. The randomization list will be generated using a computer-based randomization system (called Castor) to avoid any predictability or bias in group allocation. We apply stratified randomization to account for potential differences between the participating centers. Stratification will be based on the location of the ICU departments to ensure that any variability between sites is evenly distributed across both the intervention and control groups. This method will help mitigate confounding effects due to site-specific factors and enhance the internal validity of the study.

This study will use a double-blind approach, meaning neither the participants nor the investigators will know the treatment assignments. A matching placebo will be used to maintain blinding, ensuring that both groups receive treatments that are visually indistinguishable. Blinding will be maintained using coded study medication with the codes only being broken after data collection is complete or in the event of a medical emergency (unblinding).

14.5 Sample size, trail power and level of significance used.

As the exact standard deviation of ventilator-free days (VFD) up until day 28 is unknown for our specific population, we based our estimate on data from the PReVENT dataset (42). In this dataset, the standard deviation of VFD-28 for patients ventilated for more than 48 hours is 10.6.

Using this data, we estimate that a sample size of 250 patients (125 per group) will provide 80% statistical power to detect a difference of 3.9 ventilator-free days up until day 28. This calculation assumes a baseline standard deviation of 10.6 ventilator-free days, an alpha level of 0.05, and includes a 10% inflation rate to account for the potential non-normal distribution of VFD-28.

14.6 Planned analysis

14.6.1 Analysis primary endpoint

The primary analysis of the primary outcome, which is ventilator-free days up until day 28 (VFD-28), will reflect a composite endpoint combining both mortality and the duration of ventilation. This approach integrates the impact of death (assigned a value of 0 VFD) and the length of time a patient remains on invasive ventilation.

Day 0 will be defined as the day of randomization, with the duration of intubation prior to randomization reported as a baseline variable. The individual components of the composite endpoint—mortality rates and the duration of ventilation—will be described separately for both treatment groups to provide detailed insights into the effects of the intervention.

The primary analysis for VFD-28 will use a cumulative logistic regression model to account for the ordinal nature of the endpoint. This model will estimate odds ratios and 95% confidence intervals to quantify the effect of the treatment compared to the control group.

In addition, a sensitivity analysis will be performed to evaluate ventilator-free days at day 14 (VFD-14) and day 60 (VFD-60) to assess the robustness of the results to changes in the time window for the primary endpoint.

14.6.2 Analysis secondary endpoint(s)

- Time-to-event outcomes and mortality rates will be analysed using a cox proportional hazards regression model as described above.
- Safety endpoints such as cardiac dysrhythmias and dose of vasopressors will be analysed using linear mixed-effects models to account for repeated measures and within-subject correlations.
- Reintubation and ICU readmission will be analysed descriptively and compared between treatment groups.
- Number of days with non-invasive respiratory support (high flow nasal cannula or non-invasive ventilation) within 28 days after randomization will be analysed using descriptive statistics and compared between groups.
- Length of ICU stay will be summarized using descriptive statistics and compared between treatment groups using a linear mixed-effects model.
- EQ-5D-5L questionnaire will be summarized with descriptive statistics. Responses will be converted to utility scores using country-specific value sets. Differences in utility scores between intervention and comparator groups will be analysed using linear mixed-effects model.
- Pharmacokinetic parameters will be tabulated but not statistically evaluated, as these are descriptive exploratory parameters.

14.6.3 Analysis other study parameters/endpoints

Adverse events will be coded according to the MedDRA. All AEs noted during the trial will be listed. The absolute and relative number of patients reporting each AE by treatment group and severity will also be presented. The analysis will focus on the TEAEs which will be summarized.

A pre-planned secondary analysis will be conducted focusing on patients with cardiac failure. This analysis will specifically assess this subgroup to explore potential differential effects of treatment.

14.7 Interim analysis

No interim analysis for efficacy will be performed during this study.
Instead, we have established an independent data safety monitoring board (DSMB) to assess the safety of patients throughout the study. See section 13.11.

14.8 (Statistical) criteria for termination of the trial

The trial will be conducted until the target sample size of 250 patients is achieved. Safety and operational issues will be monitored continuously to address any potential need for early termination.

14.9 Procedure for accounting for missing, unused and spurious data

Missing data for the primary endpoint (time to successful weaning) are expected to be minimal due to continuous patient monitoring in the ICU.
For continuous or binary secondary outcomes, such as dyspnea scores or reintubation, we will use multiple imputation or mixed models to account for any missing data. Missing data will be assumed to be missing at random, allowing us to include cases with partial data in a statistically appropriate way. In the case of missing dyspnea scores, we will document the reason for each missing entry—such as if the patient was unconscious, too sick, or otherwise unable to participate—providing a clearer context for missing data patterns.

Spurious data, including outliers and anomalies, will be identified through data-cleaning procedures. Where appropriate, errors will be corrected. Outliers will be assessed and either transformed or analyzed using robust statistical methods to minimize their influence on the results.

14.10 Procedure for reporting any deviation(s) from the original statistical plan.

In case deviations from the original statistical plan are necessary, this will be documented in a note to file.

15. ETHICAL CONSIDERATIONS

15.1 Declaration of Helsinki

This study will be conducted according to the principles of the Declaration of Helsinki and in accordance with the Medical Research Involving Human Subjects Act and good clinical practice.

15.2 Recruitment and informed consent procedures

Patients admitted to the ICU will be screened for eligibility by the treating physician based on the study's inclusion and exclusion criteria.

If a patient is deemed potentially eligible, a member of the study team may approach the patient or their legal representative even before the patient undergoes a spontaneous breathing trial (SBT) to provide verbal and written information about the study. Informed consent can be obtained at this stage. If the patient subsequently fails the SBT, the study will proceed with randomization and administration of the study medication.

In cases where the patient or legal representative has not yet been informed at the time of the failed SBT, the study team will initiate the informed consent process as soon as possible after SBT failure. A window of up to 24 hours from a failed SBT is allowed to obtain informed consent, providing the legal representative or patient sufficient time to understand the study and ask questions before making a decision. Only after informed consent is obtained randomization and administration of the study medication will take place.

15.3 Benefits and risks assessment, group relatedness

If our hypothesis is correct there are several benefits. There are direct patients benefits such as

- Reduction in invasive ventilation duration. Patients may benefit from a reduction in the duration of invasive ventilation. Prolonged invasive ventilation is associated with worsened outcomes, increased risk of mortality, and extended ICU and hospital stays. By potentially shortening the duration of invasive ventilation, we aim to improve patient outcomes.
- Improved patient comfort. Invasive ventilation is often associated with severe dyspnea (9), and discomfort. This study aims to alleviate these symptoms, which are among the most distressing experiences for ICU patients.
- Enhanced communication and mobilization: reducing the need for invasive ventilation can also facilitate better communication and earlier mobilization, which are crucial for patient recovery and rehabilitation.

But apart from the direct patients benefits there is a broader impact beyond the ICU. Respiratory muscle weakness and difficulties with ventilation are common in other conditions such as chronic obstructive pulmonary disease (COPD), cancer and obesity. Insights gained from this study could inform treatment strategies for these broader patient populations.

The risks associated with the study are thoroughly outlined in chapter 4. These include potential adverse effects related to the study medication.

15.4 Compensation for injury

The sponsor has an insurance that is in accordance with the legal requirements in the Netherlands (Article 7 WMO, under 1). This insurance provides cover for damage to research subjects through injury or death caused by the study. The insurance applies to the damage that becomes apparent during the study or within 4 years after the end of the study.

- €650.000,-- (i.e. six hundred and fifty thousand Euro) for death or injury for each subject who participates in the research;
- €5.000.000,-- (i.e. five million Euro) for death or injury for all subjects who participate in the research;
- €7.500.000,-- (i.e. seven million and five hundred thousand Euro) for the total damage incurred by the organization for all damage disclosed by scientific research for the Sponsor as 'verrichter' in the meaning of said Act in each year of insurance coverage.

15.5 Compensation for subjects

There will be no compensation for subjects that will participate in this clinical trial.

15.6 Compensation for investigators

Each participating study site will receive a one-time start-up fee of €1.000,00 to cover costs related to study initiation. Besides that, sites will receive a €100,00 per patient successfully included in the study to compensate for recruitment and administrative efforts.

15.7 Other ethical considerations

Not applicable.

16. ADMINISTRATIVE ASPECTS, MONITORING AND CONFIDENTIALITY

All experiments and handling of samples will be performed according to the latest version of the declaration of Helsinki, the Medical Research Involving Human Subjects Act (WMO), and Good Clinical Practice.

16.1 Approval initial application and substantial modifications

The trial protocol, informed consent form, subject information leaflet, investigational medicinal product dossier, investigators brochure and any other documents required by the Regulation will be submitted for the regulatory approval before the clinical trial is started via CTIS.

The sponsor will also submit and obtain approval for substantial modifications to the original approved documents via CTIS.

A 'substantial modification' is defined in the CTR as any change to any aspect of the clinical trial which is made after notification of a decision referred to in Articles 8, 14, 19, 20 or 23 and which is likely to have a substantial impact on the safety or rights of the subjects or on the reliability and robustness of the data generated in the clinical trial.

16.2 Monitoring

Monitoring will be carried out by a BROK-certified internal monitor of the Radboud university medical center during the study. The stored data in the CRFs, all informed consents, SAE reports and the Trial Master File will be monitored. Monitoring will take place according to the working method, responsibilities and specific requirements described in the monitoring plan. Monitoring will be carried out by a BROK-certified internal monitor of the Radboud university medical center during the study.
In the monitor plan we classify our study as a 'minimal risk study' supported by the following arguments:

- Use of an existing medication: since levosimendan is a registered and well-researched medication, this reduces the uncertain risks associated with experimental drugs. In our pilot study with levosimendan, we did not see any serious adverse events (7).
- ICU setting: the study is conducted in a highly controlled environment with extensive monitoring. This will lead to rapid detection and treatment of any adverse events, minimizing patient risks.
- Administration of a low dose over a short intervention period.
- Establishment of a DSMB which will conduct a safety evaluation of blinded data.

16.3 Recording, handling, and storage of information

16.3.1 Handling of data and data protection

Data will be handled confidentially and coded. After informed consent, all subjects will have a unique subject code. A subject identification code list will be used to link the data to the subject. The code will not be based on the patient initials and birthdate. The key to the code will be safeguarded by the principal investigator. The handling of subject data in this study complies with the Dutch Personal Data Protection Act (in Dutch: Algemene Verordening Gegevensbescherming, AVG). Data will be collected in an electronic CRF from a validated data management system (Castor EDC). Study documents and study agreements will be kept in the Trial Master File.

16.3.2 Source documents and case report forms (CRF)

Source documents for this study will include hospital records and procedure reports and data collection forms. These documents will be used to enter data on the CRFs. Data reported on the CRF that are derived from source documents must be consistent with the source documents or the discrepancies must be explained.
All documents will be stored safely in confidential conditions. On all study-specific documents other than the signed consent, the subject will be referred to by the study subject identification code.

16.3.3 Clinical trial master file and data archiving

The sponsor and the investigator shall keep a clinical trial master file. The clinical trial master file shall always contain the essential documents relating to the clinical trial which allow verification of the conduct of a clinical trial and the quality of the data generated (CTR: Article 57).

The sponsor and the investigator shall archive the content of the clinical trial master file for at least 25 years after the end of the clinical trial unless other EU law requires archiving for a longer period. The medical files of subjects shall be archived in accordance with national law (CTR: Article 58).

The content of the clinical trial master file shall be archived in a way that ensures that it is readily available and accessible, upon request (CTR: Article 57).

16.3.4 Collection and storage of biological samples

Body materials consisting of blood will be preserved for 10 years for possible follow-up studies in a coded form. All subjects will be asked permission before preservation. If the patient does not consent to long term preservation this will not lead to exclusion within the current study. The medical ethical committee will be consulted before the body material is used for follow-up research. They will decide whether renewed subject permission is necessary.

16.4 Audits and inspections and direct access to source data/documents

This trial may be subject to internal or external monitoring, auditing, or inspections procedure to ensure adherence to GCP. Access to all trial-related documents including direct access to source data will be given at that time.

16.5 Reporting of serious breaches

The sponsor will notify the Member States concerned about a serious breach of the Regulation or of the version of the protocol applicable at the time of the breach through CTIS without undue delay but not later than **seven days** of becoming aware of that breach (CTR: Article 52).

16.6 Notification of the start and the end of the recruitment

The sponsor will notify within 15 days each Member State concerned of the start of a clinical trial in relation to that Member State through CTIS (CTR: Article 36(1)).

The sponsor will notify within 15 days each Member State concerned of the first visit of the first subject in relation to that Member State through CTIS (CTR: Article 36(2)).

The sponsor will notify within 15 days each Member State concerned of the end of the recruitment of subjects for a clinical trial in that Member State through the EU (CTR: Article 36(3)).

16.7 Temporary halt/(early) termination

The sponsor will notify within 15 days each Member State concerned of the end of a clinical trial in relation to that Member State through CTIS (CTR: Article 37(1)).

The sponsor will notify within 15 days each Member State concerned of the end of a clinical trial in all Member States concerned and in all third countries in which the clinical trial has been conducted through CTIS (CTR: Article 37(3)).

16.7.1 Temporary halt/early termination for reasons not affecting the benefit-risk balance.

The sponsor will notify with 15 days each Member State concerned of a temporary halt of a clinical trial in all Member States concerned for reasons not affecting the benefit-risk balance through CTIS (CTR: Article 37(5)).

When a temporarily halted clinical trial for reasons not affecting the benefit-risk balance is resumed the sponsor will notify each Member State concerned through CTIS (CTR: Article 37(6)).

The sponsor will notify to the EU portal CTIS of early termination of the clinical trial for reasons not affecting the benefit-risk balance through CTIS. The reasons for such action and, when appropriate, follow-up measures for the subjects will be provided as well (CTR: Article 37(7)).

16.7.2 Temporary halt/early termination for reasons of subject safety

In accordance with article 38 of the CTR, the sponsor will suspend the study if there is sufficient ground that continuation of the study will jeopardise subject health or safety. The temporary halt or early termination of a clinical trial for reasons of a change of the benefit-risk balance will be notified to the Member States concerned through the EU portal CTIS without undue delay but not later than in 15 days of the date of the temporary halt or early termination. It shall include the reasons for such action and specify follow-up measures. The restart of the clinical trial following a temporary halt as referred to in paragraph 1 shall be deemed to be a substantial modification subject to the authorisation procedure laid down in Chapter III of the CTR (CTR: Article 38).

16.8 Summary of the results

Within one year from the end of a clinical trial in all Member States concerned, the sponsor will submit to the EU database CTIS a summary of the results of the clinical trial. The content of the summary of the results is set out in CTR Annex IV. It shall be accompanied by a summary written in a manner that is understandable to laypersons. The content of the summary is set out in CTR Annex V (CTR: Article 37(4)).

16.9 Public disclosure and publication policy

The results of this study will be published in international medical journals and presented at medical conferences. In addition, a summary of the results of this study will be published on the website of study co-sponsor www.ZonMw.nl.

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